

BCA Bachelor of Computer Applications Semester -IV Theory & Practical

Teaching hours: 60/4 hrs per week

No. of Credits: 4

Course Code: BCA403

Course Title: Network Security

Subject
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Course Title : Network Security
Course Objective: Understand cryptographic terminologies, encryption methods, and cryptographic protocols, including symmetric and public-key cryptography, digital signatures, Explore intermediate and advanced cryptographic protocols such as zero-knowledge proofs, blind signatures, secure multiparty computations, digital cash etc. Learn key Management and Cryptographic Algorithm Selection. Gain knowledge of Cryptographic Foundations, Algorithms, and Real-World Implementations.

Course Outcomes

CO1: Understand Fundamentals of Cryptographic Techniques and Protocols

CO2: Analyze Advanced Cryptographic Protocols for Secure Communications

CO3: Create and develop Proficiency in Cryptographic Key Management and Encryption Techniques

CO4: Implement and Analyze cryptographic security using differential and linear cryptanalysis, pseudo-random sequence generators, and stream ciphers etc.

Employability

Entrepreneurship,

Skill development

Value enrichment,

Women empowerment

UNIT	Course Contents/Titles	Course Outcome	Cognitive Levels	Relevance L/R/N/G
I	UNIT I Cryptography Terminology, Stenography, Subscription ciphers,	CO1	Understand	G

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	one time patios, Cryptographic Protocols- Introduction using Symmetric Cryptography , one-way Hash Functions , communication using public -key cryptography and hybrid cryptosystems, Digital Signatures, Digital Signatures with encryption, Random and pseudo random sequence generation, Basic protocols- key Exchange, authentication and key exchange, formal analysis, secret splitting, secret sharing, cryptographic protection of databases.			
II	UNIT II Intermediate protocols-Time stamping, subliminal channel, Signatures, Bit commitment, fair coin flops, mental poker, key Escrow. Advanced Protocols-Zero Knowledge proofs, Blind signatures, identity based Public- key cryptography obvious transfer, simultaneous contract signing ,Digital certified main, simultaneous exchange of secrets. Esoteric Protocols-Secret Elections, secure multiparty computation, anonymous Message Broadcast Digital cash.	CO1, CO2, CO3	Understand, Analyze, Create	G
III	UNIT III Key Length-symmetric key length, public key length Key Management Generation transfer, verification use, Store Backup, lifetime, destruction,	CO1, CO2, CO3, CO4	Understand, Analyze, Create, Implement	G

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	public key management. Algorithm types and modes- Electronic codebook, Block replay, Cipher Block clouding, Stream Ciphers, self synchronizing, cipher- feedback, synchronous stream, output feedback, counter mode, choosing cipher mode, comparison Using Algorithms-choosing an algorithm, public key versus symmetric encrypting and communication channels. Encrypting data for storage hardware encryption versus software Encryption			
IV	UNIT IV Information theory, complexly theory, number theory, factoring, prime number generation DBS -Backgr ound, Descri ption, security of DES, Differential and linear crypto analysis, DB vacancy, pseudo random sequence generators, linear congruential, Linear feed back shift register stream ciphers, Design and analysis, Stream Cipher using LFSRs. One way hash functions, Background, MDS, SHA, Public- key Algorithms-Backgrounders', Digital Signature Algorithm Diffie helman algorithm for key exchange Implementation Examples-IBM, ISDN, Kerberos, ISO authentication framework, PBM PKC DEPS	CO1, CO2, CO3, CO4	Understand, Analyze, Create, Implement	G

Suggested Reading:

1. Bruce Schneier-Applied Cryptography, Wiley 2001.
2. William Stallings –Cryptography and Network Security, PH

Teaching hours: 3 hrs per week

No.of Credits: 1

Course Code: BCA413

Course Title: Network Security Lab

CO1	Develop Practical Understanding of Networking Concepts and Configurations
CO2	Gain Hands-on Experience in Network Programming and Interprocess Communication (IPC)

Networking concepts demonstration

1. Demonstrations of IP address and ports in computer system.
2. Explanation of settings in network connections
3. Testing of networking connectivity using ping, tracepath
4. Checking network statistics with netstat
5. Demonstration of static and dynamic IP address settings
6. Understanding ethernet cabling and switched networks
7. Comprehension of routers and firewalls 8. Significance of DNS

Network programming

1. Implement IPC using a) Pipes b) FIFO
2. Implement file transfer using Message Queue form of IPC
3. Design TCP iterative Client and server application to reverse the given input sentence
4. Design TCP concurrent Client and server application to reverse the given input sentence
5. Design TCP client and server application to transfer file
6. Design UDP Client and server application to reverse the given input sentence
7. Design UDP Client and server application to reverse the given input sentence
8. Design UDP Client server to transfer a file

Suggested Reading

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1. Advance UNIX Programming Richard Stevens, Second Edition Pearson Education
2. Advance UNIX Programming, N.B. Venkateswarlu, BS Publication.